

AQI PREDICTOR

Lahore Air Quality Forecasting System

Project Report

Hifza120 · 2026

github.com/Hifza120/aqi_predictor

[Live Dashboard → Streamlit App](#)

1. Project Overview

This project builds an end-to-end machine learning pipeline to forecast the Air Quality Index (AQI) for Lahore, Pakistan at 24-hour, 48-hour, and 72-hour horizons. Lahore consistently ranks among the most polluted cities in the world, making reliable air quality prediction a meaningful public health tool.

The system ingests real-time and historical air quality and weather data, engineers a rich feature set, trains and compares multiple regression models, and surfaces forecasts through a live Streamlit dashboard. Model retraining and feature ingestion are fully automated via GitHub Actions.

Key Links

[GitHub Repository](https://github.com/Hifza120/aqi_predictor) → github.com/Hifza120/aqi_predictor

[Live Streamlit App](https://aqipredictor-rygzfuyfvhp5mtzc6qeeeu.streamlit.app) → aqipredictor-rygzfuyfvhp5mtzc6qeeeu.streamlit.app

2. System Architecture

The pipeline follows a modular design with clearly separated concerns. Each component can be triggered independently or as part of an automated schedule.

fetch_data.py	Fetches live AQI + weather, pushes to Feast online store
historic_airquality.py	Backfills European AQI & pollutant data (Open-Meteo)
historic_backfill.py	Backfills hourly weather data (Open-Meteo Archive API)
feature_engineering.py	Builds lag features, rolling means, cyclical encodings, targets
feature_store_pipeline.py	Runs feature engineering and materializes to SQLite via Feast
model_training.py	Trains 5 models per horizon, picks best R^2 , saves + SHAP plots
predict.py	Loads live features from online store, runs 3-horizon forecast
dashboard.py	Streamlit app — live AQI, forecast cards, EDA charts
test.py	Validates pipeline: parquet, online store, historical pull, models

Data Sources

- AQICN API — real-time station AQI, PM2.5, PM10 for Lahore (station A471607)
 - Open-Meteo Air Quality API — hourly PM2.5, PM10, NO₂, SO₂, CO, O₃, European AQI
 - Open-Meteo Archive & Forecast APIs — temperature, humidity, wind speed, precipitation, cloud cover, boundary layer height
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3. Exploratory Data Analysis

Four years of hourly data were collected spanning June 2022 to May 2026, covering Lahore across all seasons. The dataset contains no missing values after feature engineering dropna, resulting in a clean training set.

3.1 Data Quality



Figure 1 — Missing values check. All columns are complete after feature engineering.

3.2 AQI Distribution

The AQI distribution is right-skewed with a mean of 99.5 and median of 89.0. The majority of hours fall in the Moderate band (51–100), with significant time in the Unhealthy for Sensitive Groups range.

Lahore AQI — Distribution Analysis

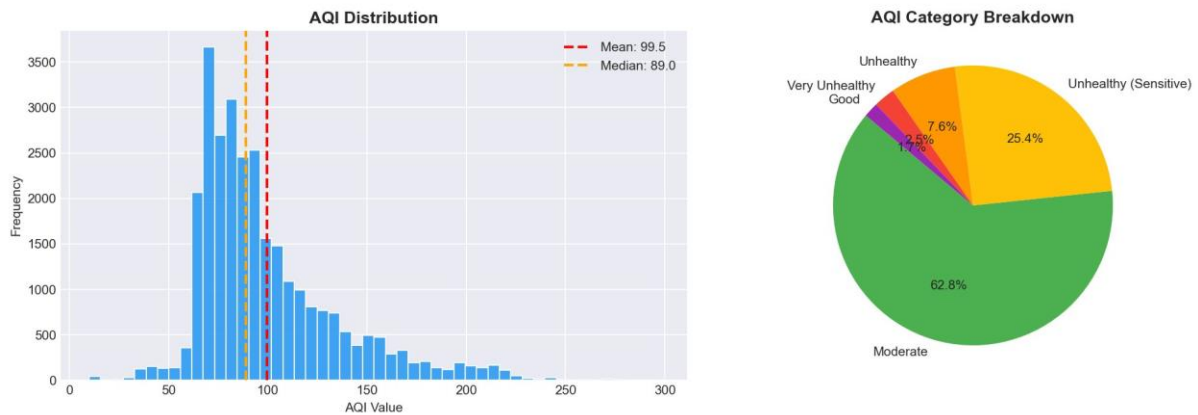


Figure 2 — AQI distribution histogram (left) and category breakdown (right). 62.8% of hours are Moderate; 25.4% Unhealthy for Sensitive Groups.

3.3 AQI Over Time

Clear seasonal cycles are visible. Winter months (October–January) show the highest AQI spikes, driven by temperature inversions that trap pollutants near ground level. Summers are relatively cleaner, though still Moderate on average.

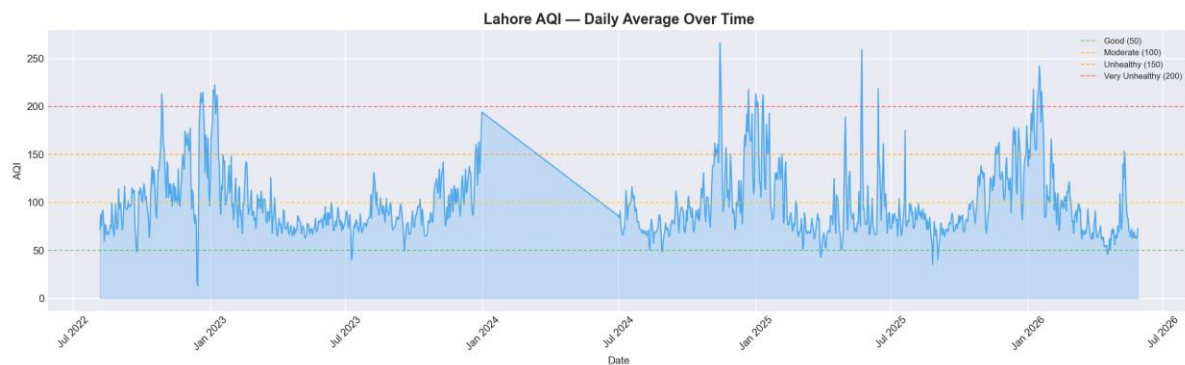


Figure 3 — Daily average AQI from July 2022 to June 2026. Threshold lines mark Good (50), Moderate (100), Unhealthy (150), and Very Unhealthy (200).

3.4 Monthly Seasonal Pattern

January is the most polluted month (avg AQI 150, Unhealthy for Sensitive Groups). April is the cleanest (avg 73, Moderate). The transition months of November and December also show elevated readings as winter smog sets in.

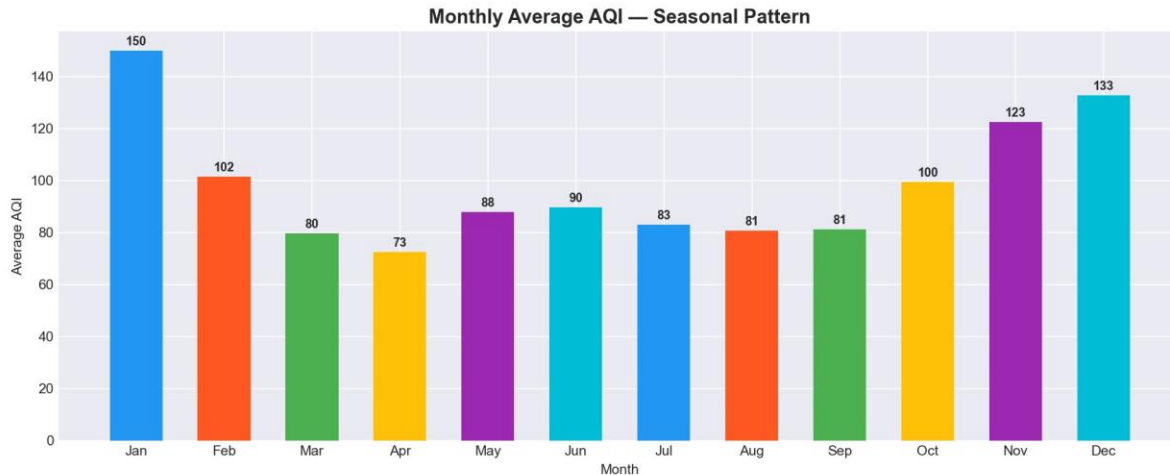


Figure 4 — Monthly average AQI. Lahore's winter smog season spans October through February.

3.5 Hourly Pattern

When measured in UTC, the hourly pattern is nearly flat across all 24 hours (range ~99–101), indicating that no single UTC hour dominates. This is expected because Lahore local time (UTC+5) shifts the diurnal cycle into different UTC bins that average out across a multi-year dataset.

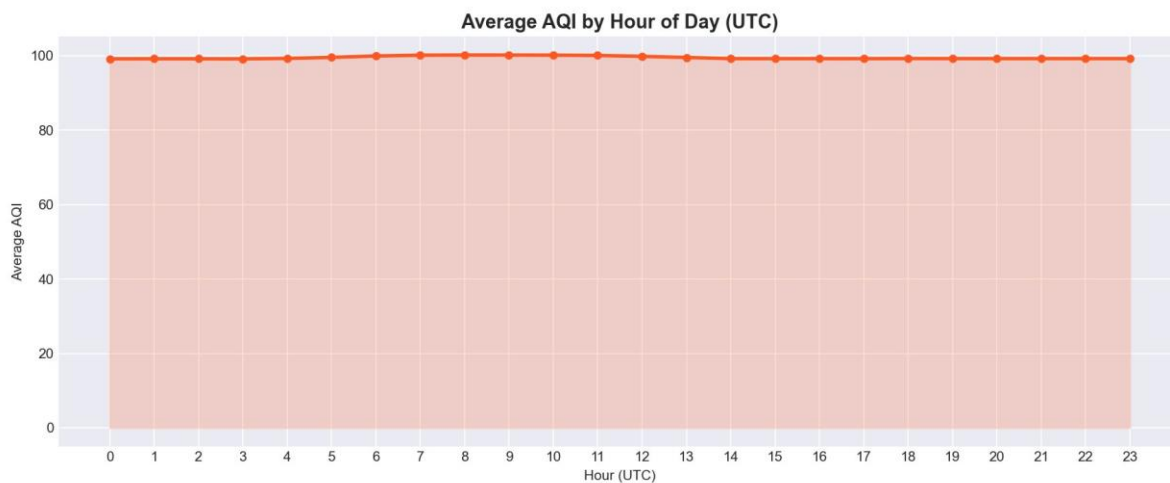


Figure 5 — Average AQI by UTC hour. The near-flat line reflects aggregation across all seasons and years.

4. Feature Analysis

4.1 Correlation Heatmap

The strongest AQI predictors are the lag and rolling-mean features of AQI and PM2.5 themselves, confirming strong temporal autocorrelation. Among pollutants, PM2.5 ($r = 0.80$) and

SO₂ ($r = 0.59$) correlate most with AQI. Temperature shows a meaningful negative correlation ($r = -0.46$), consistent with winter-trapping dynamics.

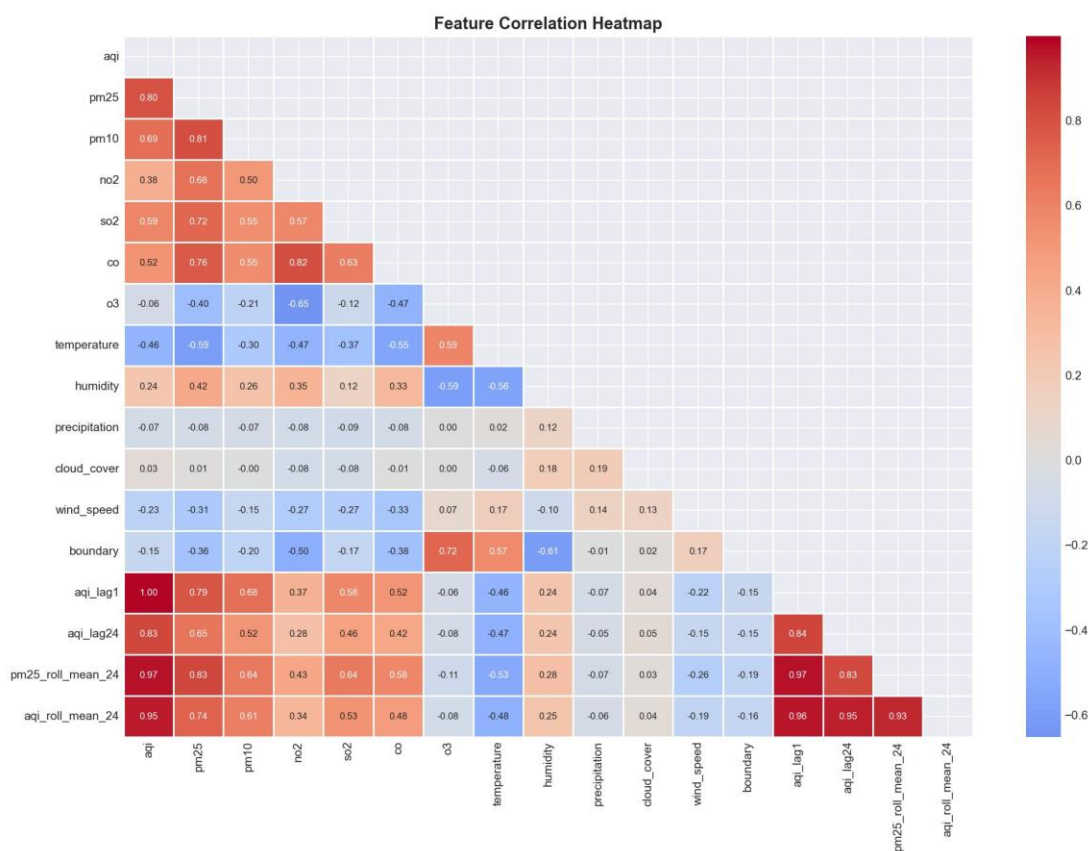


Figure 6 — Pearson correlation heatmap for selected features. *aqi_lag1* and *pm25_rol_mean_24* are near-perfect predictors of current AQI.

4.2 PM2.5 vs AQI

PM2.5 is the dominant driver of the European AQI calculation used in this dataset. The scatter plot shows a strong, non-linear relationship — AQI rises steeply as PM2.5 exceeds ~50 $\mu\text{g}/\text{m}^3$ and then plateaus at the top category thresholds.

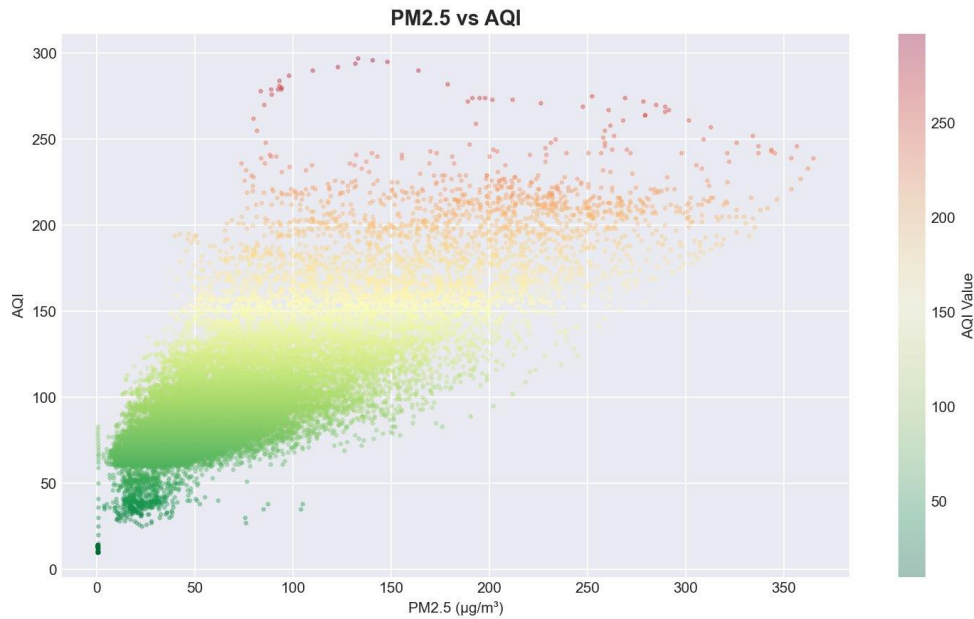


Figure 7 — PM2.5 ($\mu\text{g}/\text{m}^3$) vs AQI, coloured by AQI value. The dense cloud at low PM2.5 corresponds to the Moderate category.

4.3 Pollutant Trends

All primary pollutants exhibit the same strong winter seasonality. Carbon monoxide (CO) shows the most dramatic spikes, reaching over $7,000 \mu\text{g}/\text{m}^3$ in winter. Ozone (O_3) is counter-seasonal — highest in summer when UV radiation is strong — which explains its negative AQI correlation.



Figure 8 — Time series of PM_{2.5}, PM₁₀, NO₂, SO₂, CO, and O₃ from 2022 to 2026.

4.4 AQI vs Weather Features

Temperature shows the clearest negative relationship with AQI. Wind speed has a mild negative effect — stronger winds disperse pollutants. Precipitation shows a strong threshold effect: rain dramatically reduces AQI. Boundary layer height (atmospheric mixing depth) has a visible negative relationship; a shallower boundary traps pollution closer to the ground.

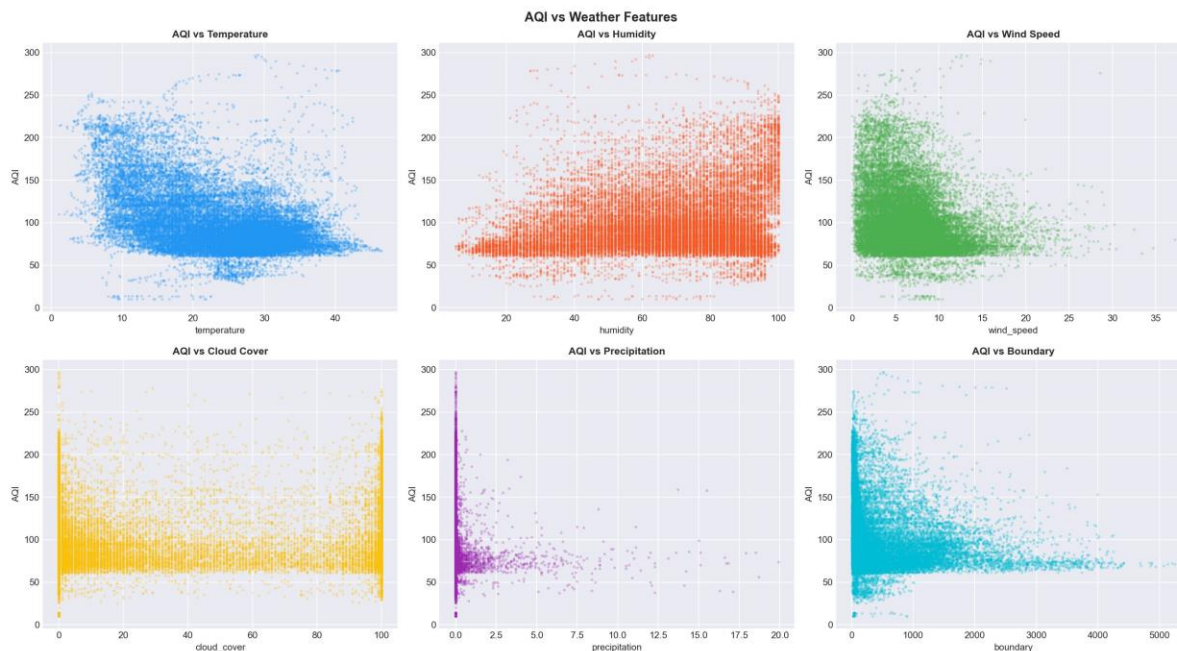


Figure 9 — Scatter plots of AQI vs temperature, humidity, wind speed, cloud cover, precipitation, and boundary layer height.

4.5 Feature Importance Ranking

Lag and rolling-mean features dominate the correlation ranking. This is expected for a forecasting problem where recent history is the best predictor of near-future air quality. The engineered cyclical features (sin/cos month, hour) contribute modestly; year shows a slight negative trend reflecting the gradual data evolution.

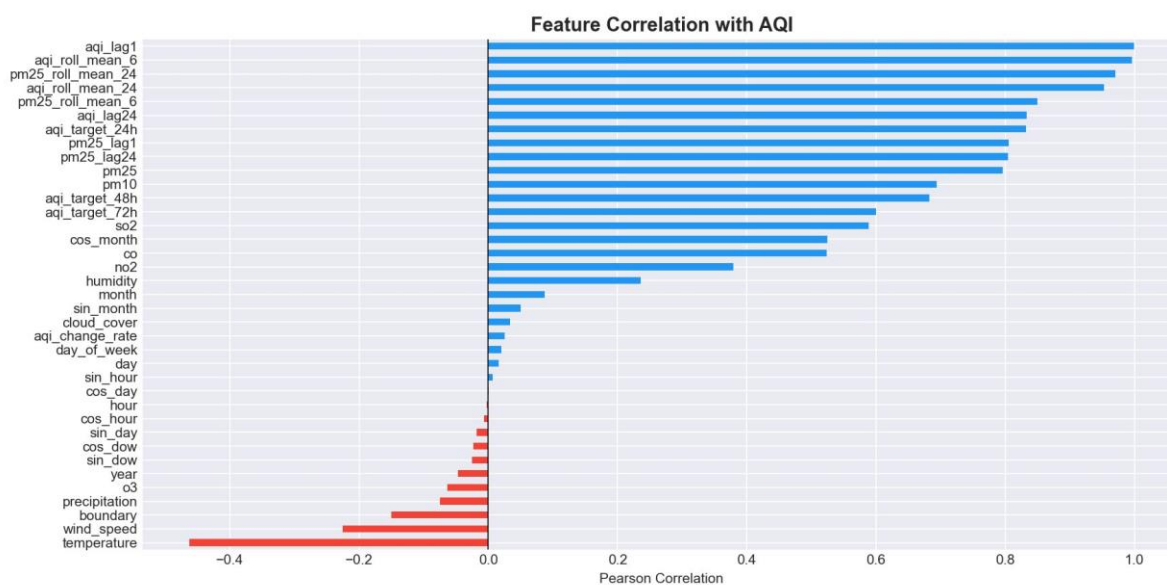


Figure 10 — Pearson correlation of all features with AQI. Blue = positive, red = negative.

5. Feature Engineering

Features are constructed in `feature_engineering.py` and stored in a Feast feature store backed by a local SQLite database. The feature set is grouped into four logical views.

Raw Features

- Weather: temperature, humidity, precipitation, cloud cover, wind speed, boundary layer height
- Pollutants: PM2.5, PM10, NO₂, SO₂, CO, O₃
- AQI from AQICN station or Open-Meteo fallback

Time Features

- Calendar: hour, day, month, year, day_of_week
- Cyclical encodings: sin/cos transforms for hour (24h cycle), month (12-month cycle), day (31-day cycle), day-of-week (7-day cycle)

Lag & Rolling Features

- PM2.5: lag 1h, lag 24h, rolling mean 6h, rolling mean 24h
- AQI: lag 1h, lag 24h, rolling mean 6h, rolling mean 24h, change rate (first difference)

Forecast Targets

- `aqi_target_24h` — AQI 24 hours ahead
- `aqi_target_48h` — AQI 48 hours ahead
- `aqi_target_72h` — AQI 72 hours ahead

Short-horizon models (24h) use the full feature set including 1-hour lags. Longer horizons (48h, 72h) drop the most recent lags that would not realistically be available at prediction time, preventing leakage.

6. Model Training

Five regression algorithms are trained and evaluated for each forecast horizon using an 80/20 chronological train/test split (`shuffle=False` to respect temporal ordering). The best model per horizon is selected by R^2 score on the held-out test set.

Algorithm	Notes
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Ridge Regression	Linear baseline with L2 regularisation
Random Forest	100 trees, n_jobs=-1
XGBoost	100 rounds, verbosity=0
Extra Trees	100 trees, n_jobs=-1
LightGBM	100 rounds, verbose=-1

Evaluation Metrics

- RMSE (Root Mean Squared Error) — penalises large errors
- MAE (Mean Absolute Error) — average absolute deviation in AQI units
- R^2 — proportion of variance explained (1.0 = perfect)

SHAP Explainability

After training, SHAP (SHapley Additive exPlanations) values are computed for the winning model at each horizon and saved as summary plots in models/. This reveals which features push individual predictions higher or lower, providing interpretability alongside predictive accuracy.

7. Automated Pipelines (GitHub Actions)

Two GitHub Actions workflows keep the system current without any manual intervention.

7.1 Hourly Feature Pipeline

Runs every hour on the scheduled cron "0 * * * *". Steps: checkout → install dependencies → write .env from secrets → feast apply → fetch_data.py → commit updated online store to main.

Trigger	Every hour (cron) + manual dispatch
Runtime	~1 min 10 sec (observed)
Output	Updated pipelines/feature_repo/data/ committed to main

7.2 Daily Training Pipeline

Runs every day at 02:00 UTC. Steps: checkout → install → write .env → feature_store_pipeline.py (full backfill + materialize) → model_training.py → commit updated models and SHAP plots.

Trigger	Daily at 02:00 UTC + manual dispatch
Output	Updated models/*.pkl and SHAP PNG plots committed to main

7.3 Workflow Run History

The screenshot below shows 13 successful workflow runs from the repository's Actions tab, confirming the hourly pipeline is operating reliably in production.

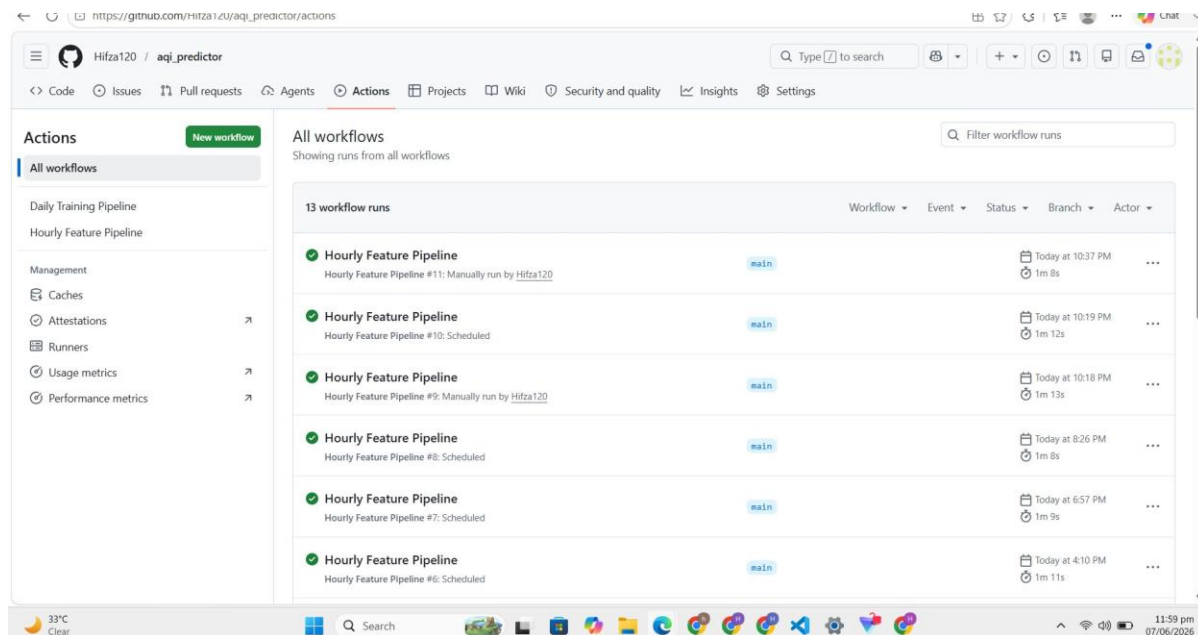


Figure 11 — GitHub Actions run history showing consecutive successful Hourly Feature Pipeline runs.

8. Live Dashboard

The forecasting system is deployed as a public Streamlit application. The dashboard displays:

- Current AQI reading and health category with colour coding
- 24h, 48h, and 72h AQI forecast cards
- Interactive EDA charts: time series, seasonal patterns, correlation heatmap
- Data sourced live from the Feast online store (SQLite, committed hourly)

Access

[Live App → aqipredictor-rygzfuyfvhp5mtzc6qeeeu.streamlit.app](https://aqipredictor-rygzfuyfvhp5mtzc6qeeeu.streamlit.app)

9. Technology Stack

Data Collection	requests, AQICN API, Open-Meteo APIs
Feature Store	Feast (local SQLite offline + online)
Feature Engineering	pandas, numpy
Visualisation (EDA)	matplotlib, seaborn
Machine Learning	scikit-learn, XGBoost, LightGBM
Explainability	SHAP
Automation	GitHub Actions (cron workflows)
Dashboard	Streamlit
Storage	Parquet (offline), SQLite (online)
Language	Python 3.11

10. Summary

This project demonstrates a production-grade MLOps pipeline for air quality forecasting tailored to one of the world's most polluted cities. Key design choices:

- Temporal train/test split — avoids data leakage and reflects real deployment conditions
- Horizon-specific feature sets — removes leaky near-term lags for longer forecasts
- Feast feature store — clean separation of offline training data and online serving
- Multi-model comparison — prevents premature commitment to any single algorithm
- Fully automated retraining — models stay current with daily fresh data
- SHAP explanations — every prediction is interpretable, not just accurate

The result is a system that runs continuously, refreshes itself, and surfaces actionable air quality forecasts for Lahore residents and researchers.